



What Makes a Level Hard?

We tried four ways to make behavior predict difficulty. Counting the pieces beat them all.

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The Problem Statement

Predict a level's human win rate from the level alone — no pre-launch player data.

- **Input:** 99 real Hex Fall levels
- **Target:** real human win rate per level (from Paxie)

What is Hex Fall?



Sort coloured pieces into matching buckets, work around ice and pin barriers, finish before you run out of room.

- slices: coloured pieces to clear
- bucket: collects one colour
- ice: frozen; thaws after N moves
- pin: barrier

First Try: RL and Why It Broke

Plan: Train an RL agent; use its win rate as the difficulty signal.

Problem: An agent learns optimal play; humans don't. optimal-player win rate $\neq$ human win rate — a structural mismatch caught before training; the simulator carried over

The Pivot

